

Class 11th: Advanced Prompting & Introduction to Web APIs

Goal: Transition into systematic automated execution loops using structured script schemas and system commands.

Phase 1: Foundation Sync (Classes 1-5 — Refresher Classes)

Focus: Reminding students of basic prompting frameworks, Claude's cloud environment, and sandbox execution.

Session	Topic	Core Learning Objectives & Details
Class 1	Re-establishing the AI Mindset	Quick recap of LLMs, ecosystem landscapes, and token context limits.
Class 2	RCTC Framework Mastery	Reviewing Role, Context, Task, and Constraints with harder data targets.
Class 3	Artifact Sandboxes Reborn	Re-engaging with no-code browser windows and rapid layout iteration.
Class 4	Introduction to Structural Tagging	Moving past plain text; introducing XML tags (, ,) to cleanly segment data.
Class 5	Conditional Logic Structures	Prompting with 'If/Then' parameters to eliminate hallucinated response loops.

Phase 2: Pipeline Automation (Classes 6-15)

Focus: Master prompt chaining setups where operational file endpoints parse outputs automatically.

Session	Topic	Core Learning Objectives & Details
Class 6	Introduction to Prompt Chaining	Breaking big problems into sequential steps (Output of Prompt A becomes Input of Prompt B).
Class 7	Automated Markdown & Document Templating	Designing repeatable templates for meeting notes, research outlines, and executive summaries.
Class 8	Designing Structured JSON Data Out	Prompting an AI to return data strictly in structured formats like JSON or CSV.
Class 9	Multi-step File Processing Pipelines	Building a workflow to ingest raw data, clean it, categorize it, and export a clean report.
Class 10	The Feedback Loop Protocol	Prompting the model to audit its own answers before showing them to the user.
Class 11	Bulk Text Summarization	Running multiple documents through an automated analysis pipeline.
Class 12	Content Repurposing Engines	Building a pipeline that turns a long-form article into slides, emails, and short updates.
Class 13	Structuring Automated Feedback Systems	Creating a prompt engine that evaluates essays or code against specific rubrics.
Class 14	Pipeline Error Handling	What to do when a prompt chain breaks midway through an execution loop.
Class 15	Pipeline Simulation Challenge	Building a functioning manual data pipeline.

Phase 3: Demystifying APIs (Classes 16-25)

Focus: Understanding the logic of data transmission headers, sending raw requests, and parsing basic backend payloads.

Session	Topic	Core Learning Objectives & Details
Class 16	What is an API?	The internet's nervous system. Connecting the cloud to external programs.
Class 17	Request vs. Response Anatomy	Demystifying URLs, Methods (GET, POST), and JSON payloads.
Class 18	API Authentication Security	Understanding API Keys, transmission headers, and cloud security rules.
Class 19	The Claude/LLM API Endpoint	Peeking under the hood of how programs send prompts to Claude directly without using the web browser.
Class 20	Testing Endpoints in Postman/ Browser	Sending your first live request and analyzing the raw JSON response payload.
Class 21	Extracting and Parsing the Payload	Writing simple text filters to grab exactly the data you need from a messy response.
Class 22	Connecting Public Data APIs	Pulling live weather, stock, or news data into text streams.
Class 23	Merging AI Prompts with Live API Data	Feeding live API data into your prompt chains for dynamic reasoning.
Class 24	Rate Limits & Token Controls	Managing budget, efficiency, and request throttling in the cloud.
Class 25	API Logic Review Quiz & Build	A team challenge connecting an external data source to a structured prompt.

Phase 4: Automation Capstone (Classes 26-35)

Focus: Building a script automation module that automatically monitors input data records and structures files cleanly.

Session	Topic	Core Learning Objectives & Details
Class 26	Automation Capstone Blueprinting	Designing a system that watches for incoming text/files, processes them via AI, and logs the result.
Class 27	Setting up the Input Trigger	Designing the ingestion step for new file/data structures.
Class 28	Integrating the Prompt Automation Chain	Embedding the XML/JSON processing templates built in Phase 2.
Class 29	Connecting the API Transmission Layer	Implementing the web logic learned in Phase 3.
Class 30	Data Cleanliness Control	Ensuring the script structures output files flawlessly without human intervention.
Class 31	Building the Notification UI	Giving your automation pipeline a simple status interface (success/fail alerts).
Class 32	Edge Case Debugging	Fixing issues when empty files or corrupted records enter the automation script.
Class 33	Optimization Overhaul	Speeding up the pipeline and reducing API token waste.
Class 34	Live Deployment & System Testing	Running the complete automation script over large datasets.
Class 35	Automation Demo Day	Students present their silent, hands-off cloud automation systems.